"Reevaluating Agricultural Productivity Gaps with Longitudinal Microdata"

Readme File for Replication Materials

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August 2020

This readme file explains how to use the files provided on Harvard's Dataverse to replicate the analysis presented in "Reevaluating Agricultural Productivity Gaps with Longitudinal Microdata", by Joan Hamory, Marieke Kleemans, Nicholas Y. Li, and Edward Miguel, published in the *Journal of the European Economic Association*. The replication package contains seven Stata do-files, two datasets (one for Indonesia and one for Kenya), all tables and figures presented in the main paper as well as in the appendix, and a log file. Section 1 of this document describes the data files, Section 2 explains the structure of the Stata do-files provided, and Section 3 discusses the remainder of the replication package.

Any questions related to these replication files can be directed to Marieke Kleemans at kleemans@illinois.edu.

1. Data

For the Indonesian portion of the analysis, we employ variables constructed from the Indonesian Family Life Survey (henceforth IFLS). The IFLS is a continuous longitudinal survey that is composed of five waves, conducted in 1993, 1997-1998, 2000, 2007-2008 and 2014-2015. The sample is representative of 83% of the country's population (Strauss, Witoelar and Sikoki, 2016). The raw IFLS datasets, questionnaires, codebooks, and User's Guides are made available by the RAND Corporation: https://www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS.html. Accessing the data require previous registration on the RAND

website, after which all data and supporting documents can be freely downloaded. The replication provided here includes the data file "IFLS.dta," which contains the IFLS data, with rounds merged together and measures used in the paper constructed. This data file is needed to run the do-files discussed in Section 2.

The Kenyan data that we use in the analysis come from the Kenya Life Panel Survey (henceforth KLPS), which tracks 8,999 individuals who attended primary school in western Kenya in the late 1990s and early 2000s over a 15-year period. The first round of the KLPS (collected during 2003-2007) can be downloaded here. Additional rounds of KLPS data (collected during 2007-2009, and during 2011-2014) will be made available in raw form in the near future, and will be posted in the KLPS Dataverse. Accessing the KLPS data on Dataverse requires registration, after which the raw data files and accompanying documentation can be freely downloaded. The replication provided here includes the data file "KLPS.dta", which contains data from KLPS Rounds 1, 2, and 3, with rounds merged together and measures used in the paper constructed. This data file is needed to run the do-files discussed in Section 2

Please refer to Section 3 of the paper for further details about the data.

2. Stata do-files

This replication package consists of the following 7 Stata do-files. Comments and explanations are included in the files.

Individuals seeking to replicate the analysis in the paper will need to save all data and do-files in a single folder and run the master do-file to obtain the figures and tables in the paper and accompanying appendix.

- Do-file "**1. Run_HKLM.do**" is the master do-file that runs all other do-files.
- Do-file "**2. Creation of IFLS replication data.do**" creates three main datasets that are used in the do-file "3. Replication analysis IFLS.do":
    - Main_Analysis_IFLS.dta
    - Consumption_Analysis_IFLS.dta
    - Intergen_Analysis_IFLS.dta
- Do-file "**3. Replication analysis IFLS.do**" uses the three data files created by the previous do-files and creates all figures and tables that use data from Indonesia.
- Do-file "**4. Creation of KLPS replication data.do**" creates four main datasets that are used in the do-file "5. Replication analysis KLPS.do":
    - Main_Analysis_KLPS.dta
    - Meals_Analysis_KLPS.dta
    - Consumption_Analysis_KLPS.dta
    - Intergen_Analysis_KLPS.dta
- Do-file "**5. Replication analysis KLPS.do**" uses the three data files created by the previous do-files and creates all figures and tables that use data from Kenya.
- "**6. Cr23_ik_css_mod.do**" is an auxiliary do-file used to create Appendix Table 16 "Alternative Coefficient Standard Error Estimation." It is a lightly modified version of cr23_ik_css.do found in the replication package for the paper "A Practitioner's Guide to Cluster-Robust Inference", by A. Colin Cameron and Douglas L. Miller, published in the Journal of Human Resources, 2015. The original version of this do file can be downloaded here.

- "**7. Edfreg_mod.do**" is an auxiliary do-file used to create Appendix Table 16 "Alternative Coefficient Standard Error Estimation." It is a lightly modified version of edfreg.ado made available by Alwyn Young. The original version of this file can be downloaded here.

3. Output and log files

In addition to the files needed to create all tables and figures, this replication package also includes a zip file with all tables and figures as they appear in the paper. Tables 1, 2 and 3 below list all tables and figures together with the name of the files created using the Stata do-files. Finally, this replication package includes the Stata log file "Log_HLKM.txt" created when running the master do-file "Run_HKLM.do" that starts all other code.

Table 1. Overview of Tables and Figures in the Main Text

Table/Figure	Panel	Output file
Figure 1	Panel A	Figure1_IFLS_agnonag.gph
	Panel B	Figure1_KLPS_agnonag.gph
	Panel C	Figure1_IFLS_ruralurban.gph
	Panel D	Figure1_KLPS_ruralurban.gph
	Combined graph	Figure1.eps
Figure 2	Panel A	Figure2_IFLS.jpg
	Panel B	Figure2_KLPS.jpg
Figure 3	Panel A	Figure3_IFLS.eps
	Panel B	Figure3_KLPS.eps
Table 1	Panel A	Table1_IFLS.tex
	Panel B	Table1_KLPS_PanelB.tex
	Panel C	Table1_KLPS_PanelC.tex
Table 2	Panel A	Table2_IFLS.tex
	Panel B	Table2_KLPS.tex
Table 3	-	Table3.tex
Table 4	Panel A	Table4_IFLS.tex
	Panel B	Table4_KLPS.tex
Table 5	Panel A	Table5_IFLS.tex
	Panel B	Table5_KLPS.tex
Table 6	Panel A	Table6_IFLS_rural.tex
	Panel B	Table6_IFLS_urban.tex

Table 2. Overview of Figures in the Appendix

Figure	Panel	Output file
Appendix Figure A1	-	FigureA1.pdf
Appendix Figure A2	-	FigureA2.pdf
Appendix Figure A3	-	FigureA3.pdf
Appendix Figure A4	Panel A	FigureA4_IFLS.eps
	Panel B	FigureA4_KLPS.eps
Appendix Figure A5	-	FigureA5_IFLS.eps
Appendix Figure A6	-	FigureA6_IFLS.eps
Appendix Figure A7	Panel A	FigureA7_IFLS.eps
	Panel B	FigureA7_KLPS.eps
Appendix Figure A8	Panel A	FigureA8_IFLS_rural.eps
	Panel B	FigureA8_IFLS_urban.eps
	Panel C	FigureA8_KLPS_rural.eps

Table 3. Overview of Tables in the Appendix

Table	Panel	Output file
Appendix Table A1	Panel A	TableA1_IFLS.tex
	Panel B	TableA1_KLPS.tex
Appendix Table A2	Panel A	TableA2_IFLS.tex
	Panel B	TableA2_KLPS.tex
Appendix Table A3	-	TableA3_IFLS.tex
Appendix Table A4	-	TableA4_IFLS.tex
Appendix Table A5	-	TableA5_KLPS.tex
Appendix Table A6	Panel A	TableA6_IFLS.tex
	Panel B	TableA6_KLPS.tex
Appendix Table A7	Panel A	TableA7_IFLS.tex
	Panel B	TableA7_KLPS.tex
Appendix Table A8	Panel A	TableA8_IFLS.tex
	Panel B	TableA8_KLPS.tex
Appendix Table A9	Panel A	TableA9_IFLS.tex
	Panel B	TableA9_KLPS.tex
Appendix Table A10	Panel A	TableA10_IFLS.tex
	Panel B	TableA10_KLPS.tex
Appendix Table A11	-	TableA11_IFLS.tex
Appendix Table A12	Panel A	TableA12_IFLS.tex
	Panel B	TableA12_KLPS.tex
Appendix Table A13	Panel A	TableA13_IFLS.tex
	Panel B	TableA13_KLPS.tex
Appendix Table A14	Panel A	TableA14_KLPS_PanelA.tex
	Panel B	TableA14_KLPS_PanelB.tex
Appendix Table A15	Panel A	TableA15_KLPS_unemp.tex
	Panel B	TableA15_KLPS_search.tex
Appendix Table A16	Panel A	TableA16_IFLS.tex
	Panel B	TableA16_KLPS.tex
Appendix Table A17	Panel A	TableA17_IFLS.tex
	Panel B	TableA17_KLPS.tex
Appendix Table A18	-	TableA18_KLPS.tex
Appendix Table A19	Panel A	TableA19_IFLS.tex
	Panel B	TableA19_KLPS.tex
Appendix Table A20	Panel A	TableA20_IFLS_food.tex
	Panel B	TableA20_IFLS_nfood.tex
Appendix Table A21	Panel A	TableA21_IFLS_tot.tex
	Panel B	TableA21_IFLS_food.tex
	Panel C	TableA21_IFLS_nfood.tex
Appendix Table A22	Panel A	TableA22_IFLS_rural.tex
	Panel B	TableA22_IFLS_urban.tex
Appendix Table A23	Panel A	TableA23_IFLS.tex
	Panel B	TableA23_KLPS.tex